

Avera - Track 1 Ranking Engine

Redrob India Runs ? Intelligent Candidate Discovery

The trap Redrob built into the dataset

"The right answer is not find candidates whose skills section contains the most AI keywords."

- Marketing managers with perfect ML skill lists
- Fictional companies (60% of pool)
- Behavioral ghosts - perfect on paper, unavailable in practice

Our thesis

Hybrid semantic + deterministic ranking

- Embeddings capture career narrative fit
- Rules enforce JD constraints (YOE, cities, honeypots)
- Behavioral multiplier for hireability

Pipeline

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100K JSONL -> validate -> filter fiction -> detect honeypots

-> score (5 dimensions) -> behavioral x -> heap top-100 -> reasoning -> CSV

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$O(N \log K)$ memory ? Streaming ingest ? CPU-only

Scorer weights (JD-backed)

Title & career | 35%

Skills | 25%

Semantic (MiniLM) | 15%

Experience | 15%

Location | 10%

Behavioral | x0.4-x1.3

Semantic layer (ADR-003)

Model: all-MiniLM-L6-v2

Candidate text: headline + summary + career_history descriptions

Not used: raw skill keyword density (honeypot vector)

Honeypot defense

1. Fictional company pre-filter
2. Title / skill mismatch
3. Expert-with-zero-months anomaly
4. Impossible seniority
5. Unverified generalist

DevOps / SRE signals

- GitHub Actions: lint ? test ? security ? integration
- Makefile one-command CI
- Structured JSON logs + run_id
- defusedcsv ? Docker ? HF sandbox
- Runbook for Day-2 ops

Reproduce in one command

```
`bash
```

```
python rank.py --candidates DataSet/candidates.jsonl --out submission.csv
```

```
python DataSet/validate_submission.py submission.csv
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Offline: make download-model -> set AVERA_SEMANTIC_MODEL

Why us for Redrob

- Thinks like Senior AI Engineer (embeddings, ranking theory, honeypot reasoning)
- Ships like DevOps/SRE (CI, observability, security, reproducibility)
- Built for India scale - 100K in minutes on CPU

Repo: github.com/GunaPalanivel/Avera

Sandbox: huggingface.co/spaces/GunaPalanivel/avera-ranker

PDF deck: docs/submission/deck.pdf - regenerate with make export-pdf.